Multi-Gene Biomarkers Results:

Correlation of the genetic findings with the clinical condition of the patient is required to arrive at accurate diagnosis, prognosis or therapeutic decisions.

Test Description

Liquidseq Actionable Genomic Profiling Panel is advancing precision oncology research through the analysis of multiple relevant biomarkers in a single next-generation sequencing (NGS) test.

The Liquidseq Actionable Genomic Profiling assay analyses 34 unique genes (28 DNA + 6 Fusions driver genes) for single gene and multiple gene biomarker insights, detects single gene biomarkers from all variant types including SNVs, indels, CNVs, known and novel fusions, and splice variants. Analyze complex multi-gene biomarkers for mutational signatures, including microsatellite instability (MSI) for immunotherapy research.

The performance of the panel was verified using both commercially sourced reference controls and cell-free DNA (cfTNA)samples.

Test Methodology Extraction & Library Preparation: Cell free Total Nucleic acid (cfTNA) extracted from the liquid biopsy sample is used for targeted gene capture using a custom designed hybrid capture panel. In process quality controls were determined for prepared library.

References

Key Findings: *Genetic test results are reported based on the recommendations of AMP-ASCO-CAP guidelines. **Public data sources included in relevant targeted therapies, guidelines: EMA, FDA, ESMO, NCCN.

Notes Treatment decisions should not be based solely on the results of this test. Findings must be interpreted in the context of the patient's complete clinical profile—including pathological, radiological, and family history—to guide diagnosis, prognosis, and therapeutic decisions. Therapy-related information in this report is derived from multiple sources including FDA-approved drug data, NCCN guidelines, peer-reviewed literature, standard clinical databases, and biomarker evidence strength. However, therapy options listed may not be suitable or beneficial for all patients. This report aims to provide a summary of potentially effective treatments, therapies that may pose increased risks, and those that may have limited benefit, based on genomic alterations identified. Prognostic and diagnostic biomarkers relevant to the disease context may also be reported. Laboratory does not guarantee completeness, accuracy, or eligibility for trial enrollment. Physicians must verify eligibility criteria for specific trials independently. Identification of a genomic alteration does not guarantee therapeutic efficacy or predict ineffectiveness. Final decisions on treatment should rest with the treating physician. The absence of reported therapeutic options does not imply ineffectiveness or lack of safety of any medication selected independently. All variant classifications and related clinical information are based on the latest evidence from peer-reviewed publications, public clinical databases, and medical guidelines (e.g., NCCN, ASCO, AMP) available at the time of report generation. However, literature is continuously evolving, and classification may change with emerging evidence. The assay is designed and validated for somatic variant detection, it does not differentiate between somatic and germline variants. If indicated, germline testing and genetic counselling are recommended.

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Table 1 (Page 1)

Tier I	Tier II	Tier III	MSI	Therapies	Clinical Trials
0	1	0	**Stable**	0	43

Table 2 (Page 1)

Tier	Gene & Transcript	Variant	Exon	Coverage/VAF
IIC	TP53 NM 000546.6 _	c.405C>A p.(C135*)	5	691x / 93.13%

Table 3 (Page 1)

Assay Biomarker	Result	Interpretation
Microsatellite Instability (MSI)	**Stable**	**Stable Microsatellite detected**

Table 4 (Page 2)

Assay Biomarker	Result	Interpretation
Variant	Therapy	
	In This Cancer	In Other Cancer
IIC TP53 p.(C135*)	▪ None*	▪ None*

Table 5 (Page 2)

Variant	Diagnostic Significance	Prognostic Significance
TP53 p.(C135*)	None*	None*

Table 6 (Page 2)

Variant	Biomarker Description
TP53 NM 000546.6 _ chr17:7578525 c.405C>A p.(C135*)	**The TP53 gene encodes the tumor suppressor protein p53, which binds to DNA and activates transcription in response to diverse cellular stresses to induce cell cycle arrest, apoptosis, or DNA repair[1]. In unstressed cells, TP53 is kept inactive by targeted degradation via MDM2, a substrate recognition factor for ubiquitin-dependent proteolysis[2]. Alterations in TP53 are required for oncogenesis as they result in loss of protein function and gain of transforming potential[3]. Germline mutations in TP53 are the underlying cause of Li-Fraumeni syndrome, a complex hereditary cancer predisposition disorder associated with early-onset cancers[4,5]. TP53 is the most frequently mutated gene in the cancer genome with approximately half of all cancers experiencing TP53 mutations. Ovarian, head and neck, esophageal, and lung squamous cancers have particularly high TP53 mutation rates (60-90%)[6,7,8,9,10,11]. Approximately two-thirds of TP53 mutations are missense mutations and several recurrent missense mutations are common, including substitutions at codons R158, R175, Y220, R248, R273, and R282[6,7]. Invariably, recurrent missense mutations in TP53 inactivate its ability to bind DNA and activate transcription of target genes[12,13,14,15]. Alterations in TP53 are also observed in pediatric cancers[6,7]. Somatic mutations are observed in 53% of non-Hodgkin lymphoma, 24% of soft tissue sarcoma, 19% of glioma, 13% of bone cancer, 9% of B-lymphoblastic leukemia/lymphoma, 4% of embryonal tumors, 3% of Wilms tumor and leukemia, 2% of T-lymphoblastic leukemia/lymphoma, and less than 1% of peripheral nervous system cancers (5 in 1158 cases)[6,7]. Biallelic loss of TP53 is observed in 10% of bone cancer, 2% of Wilms**

Table 7 (Page 3)

Variant	Biomarker Description
	tumor, and less than 1% of B-lymphoblastic leukemia/lymphoma (2 in 731 cases) and leukemia (1 in 250 cases)[6,7]. The small molecule p53 reactivator, PC14586[16] (2020), received a fast track designation by the FDA for advanced tumors harboring a TP53 Y220C mutation. The FDA has granted fast track designation to the p53 reactivator, eprenetapopt[17], (2019) and breakthrough designation[18] (2020) in combination with azacitidine or azacitidine and venetoclax for acute myeloid leukemia patients (AML) and myelodysplastic syndrome (MDS) harboring a TP53 mutation, respectively. In addition to investigational therapies aimed at restoring wild-type TP53 activity, compounds that induce synthetic lethality are also under clinical evaluation[19,20]. TP53 mutation are a diagnostic marker of SHH-activated, TP53-mutant medulloblastoma[21]. TP53 mutations confer poor prognosis and poor risk in multiple blood cancers including AML, MDS, myeloproliferative neoplasms (MPN), and chronic lymphocytic leukemia (CLL), and acute lymphoblastic leukemia (ALL)[22,23,24,25,26,27]. In mantle cell lymphoma, TP53 mutations are associated with poor prognosis when treated with conventional therapy including hematopoietic cell transplant[28]. Mono- and bi-allelic mutations in TP53 confer unique characteristics in MDS, with multi-hit patients also experiencing associations with complex karyotype, few co-occurring mutations, and high-risk disease presentation as well as predicted death and leukemic transformation independent of the IPSS-R staging system[29].

Table 8 (Page 3)

Therapy	FDA	NCCN	EMA	ESMO	Trials
niraparib	X	X	X	X	II
atezolizumab	X	X	X	X	II
azenosertib	X	X	X	X	II
ART-6043, olaparib, niraparib	X	X	X	X	I/II
ATRN-119	X	X	X	X	I/II

Therapy	FDA	NCCN	EMA	ESMO	Trials
niraparib, GSK-101	X	X	X	X	I/II
VB-15010	X	X	X	X	I/II
GenSci-122 Pag	X e 3 of 10	X	X	X	I

Table 9 (Page 4)

Therapy	FDA	NCCN	EMA	ESMO	Trials
GH-2616	X	X	X	X	I
HRS-1167	X	X	X	X	I
HS-10502	X	X	X	X	I
SIM-0501	X	X	X	X	I
talazoparib, palbociclib, axitinib, crizotinib	X	X	X	X	I
XL-309, olaparib	X	X	X	X	I
pembrolizumab, chemotherapy	X	X	X	X	IV
hormone therapy, chemotherapy	X	X	X	X	III
olaparib, chemotherapy, radiation therapy	X	X	X	X	III
saruparib, hormone therapy	X	X	X	X	III
niraparib, radiation therapy	X	X	X	X	II/III
olaparib + chemotherapy	X	X	X	X	II/III
abiraterone + niraparib, hormone therapy, radiation therapy, steroid	X	X	X	X	II
camrelizumab, fluzoparib, chemotherapy	X	X	X	X	II
eflornithine	X	X	X	X	II
hormone therapy, hormone therapy + chemotherapy	X	X	X	X	II
lunresertib, trastuzumab, chemotherapy, camonsertib	X	X	X	X	II
niraparib, dostarlimab	X	X	X	X	II
olaparib	X	X	X	X	II
olaparib + hormone therapy	X	X	X	X	II
olaparib, chemotherapy	X	X	X	X	II

Table 10 (Page 5)

Therapy	FDA	NCCN	EMA	ESMO	Trials
olaparib, durvalumab	X	X	X	X	II
olaparib, pembrolizumab	X	X	X	X	II
sintilimab, chemotherapy	X	X	X	X	II
fluzoparib, camrelizumab, chemotherapy	X	X	X	X	I/II
IMP-1734	X	X	X	X	I/II
tislelizumab, radiation therapy	X	X	X	X	I/II
A2A-252	X	X	X	X	I
GB-3010	X	X	X	X	I
lunresertib, camonsertib, Debio-0123	X	X	X	X	I
SC0245	X	X	X	X	I

Table 11 (Page 7)

DNA Gene List				
AKT1	ALK	APC	ATM	BRAF
BRCA1	BRCA2	CTNNB1	EGFR	ERBB2
ERBB3	ESR1	FOXL2	GNA11	GNAQ
HRAS	IDH1	IDH2	KIT	KRAS
MET	NRAS	PDGFRA	PIK3CA	POLE
RAF1	RET	TP53		

Table 12 (Page 7)

RNA Gene List				
ALK	MET	NTRK1	NTRK2	NTRK3
ROS1				

Table 13 (Page 7)

MSI	% of Unstable Loci
MSI - Stable	0-19%
MSI - Low / Intermediate	20-30%
MSI - High	>30%

Table 14 (Page 7)

Tier	Interpretation
Tier IA	**Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to US FDA- approved therapies for a specific type of tumor or have been included in professional guidelines as therapeutic, diagnostic, and/or prognostic biomarkers for specific types of tumors.**
Tier IB	Variants of Strong Clinical Significance: Biomarkers that predict response or resistance to a therapy based on well-powered studies with consensus from experts in the field, or have diagnostic and/or

Table 15 (Page 8)

Tier	Interpretation
	prognostic significance of certain diseases based on well-powered studies with expert consensus.
Tier IIC	**Variants of Potential Clinical Significance: biomarkers that predict response or resistance to therapies approved by FDA or professional societies for a different tumor type (i.e., off-label use of a drug), serve as inclusion criteria for clinical trials, or have diagnostic and/or prognostic significance based on the results of multiple small studies.**
Tier IID	Variants of Potential Clinical Significance: Biomarkers that show plausible therapeutic significance based on preclinical studies or may assist disease diagnosis and/or prognosis themselves or along with other biomarkers based on small studies or multiple case reports with no consensus.
Tier III	**Variants of Unknown clinical significance (VUS): Not observed at a significant allele frequency in the general or specific subpopulation or pancancer or tumor specific variant databases. No convincing published evidence of cancer Association**

Extracted Body Images & Graphical Content

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